Automatic Wildlife Monitoring System

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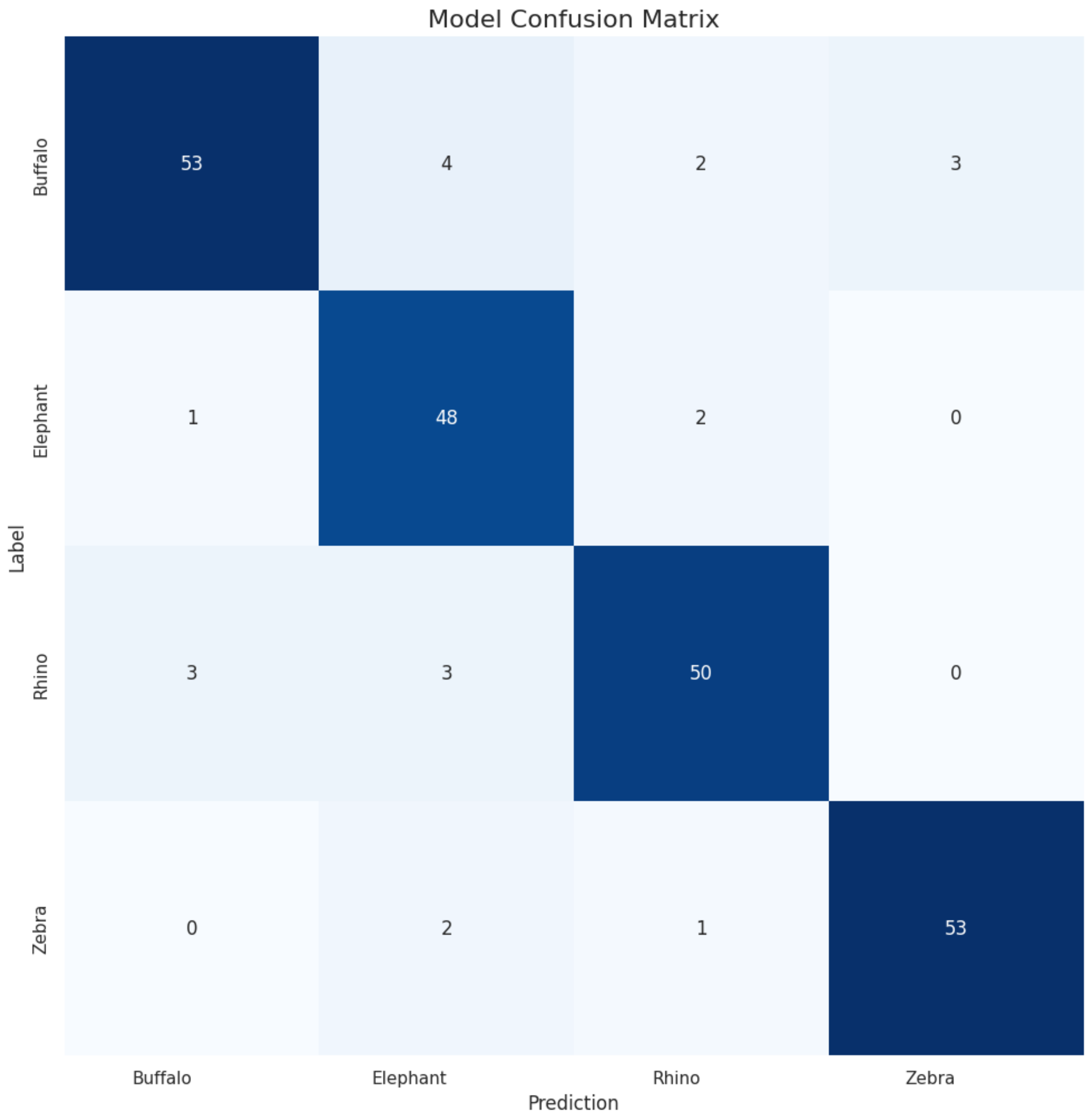
Problem & Motivation

- **Problem Context:** Endangered species, face severe threats from poaching, habitat loss, and human-wildlife conflict..
- **Research Gap:** Existing monitoring methods lack real-time automated identification capabilities and efficient tracking mechanisms suited for rugged local environments.
- **Objective:** To develop and evaluate a robust, real-time, automated wildlife detection and classification system for Performance optimization.

Evaluation & Key Results

Performance

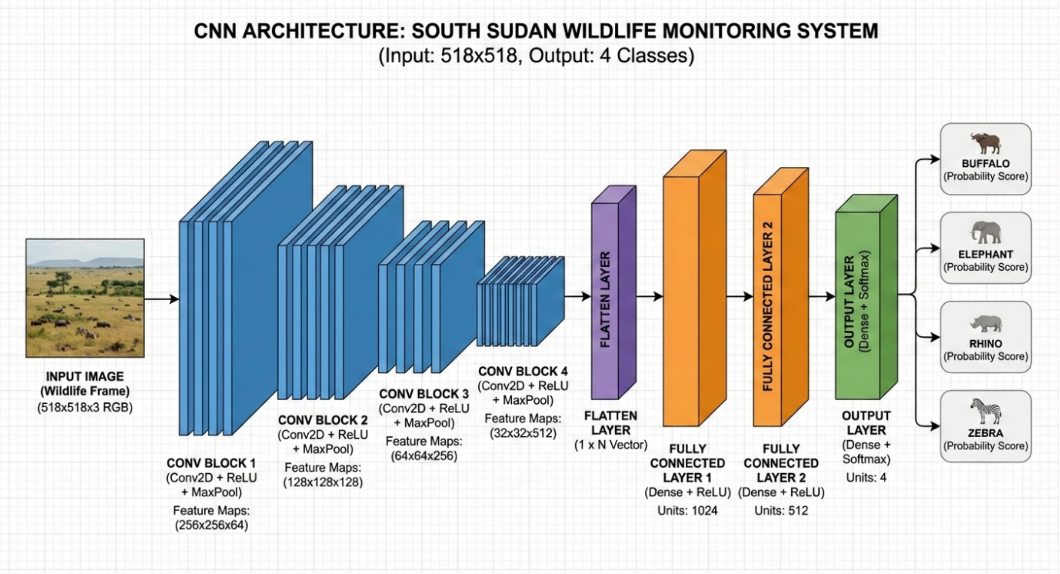
*We evaluated the Convolutional Neural Network (CNN) model on a held-out test set and achieved **90.7% accuracy** [with an overall F1-score of 0.91].*



Methodology & Dataset

System Architecture

Our system utilizes a CNN to classification images into four categories (Buffalo, Elephant, Rhino, and Zebra)



Dataset

Our data is comprised of over 1,500 annotated images sourced from local camera traps, drone footage, and open-source African wildlife repositories, focusing on four target species: Elephant, Buffalo, Rhino, and Zebra

Significance & Conclusion

- **Contribution:** A robust deep learning framework tailored for automated wildlife tracking and classification of key endangered species in South Sudan
- **Impact:** Can be deployed via edge computing devices and camera trap networks to assist conservationists and park rangers in real-time anti-poaching operations.
- **Future Work:** Future work would focus on moving beyond static image classification to Temporal Action Recognition by analyzing video frames instead of single images .